Exploratory Analysis of the Chromosome 13 on DNA Repeat Composition and Functional Disease Related Genes

Jhon Esterilla Viafara1*, Sebastian Cuadros Mateus1, Diego Gómez Ortega1 and Fabian Tobar-Tosse2*

1. Medical school, Pontificia Universidad Javeriana Cali, Cali 760031, Colombia
2. Department of Basic Sciences for Health, Pontificia Universidad Javeriana Cali, Cali 760031, Colombia

Pontificia Universidad JAVERIANA Cali

iSCB INTERNATIONAL SOCIETY FOR COMPUTATIONAL BIOLOGY

BACKGROUND

The study of human chromosomes is pivotal to understanding the genetic basis of diseases. Research into repetitive DNA sequences has advanced our knowledge of gene regulation, expression, and disease susceptibility. Studying the repetitive DNA composition of chromosome 13 (Chr13) will allow a better understanding of the genetic mechanisms underlying Chr13-associated diseases, providing deeper insights into the role of chr13 within the human genome.

MATERIALS AND METHODS

This study examines the landscape of repetitive DNA elements in Chromosome 13 (Chr13) to identify potential patterns linked to pathological processes. Using bioinformatic approaches, we aim to elucidate molecular interactions among Chr13 genes, their known disease associations, and the chromosome's unique genomic features.

- *Genomic Information Retrieval of Chromosome 13*
- *Functional Enrichment Analysis*
- *Protein Interaction Definition and Related Processes*
- *Identification of Associated Diseases*

UCSC Genome Browser · NCBI · Enrichr · STRING · ShinyGO · GO the Gene Ontology

RESULTS AND DISCUSSION

The research has revealed that chromosome 13 harbors genes associated in literature with a with a diverse range of diseases, including neurodegenerative, neuromuscular, gastrointestinal, metabolic, immunohematological, and dermatological conditions.

The complexity of interactions among chromosome 13 genes and the lack of direct associations with specific disease processes have posed challenges in establishing significant connections due to the statistical differences between the bioinformatic databases.

The relatively low pathogenic element load on chromosome 13 provides a unique opportunity to investigate the impact of its plasticity. This impact extends beyond aneuploidy or chromosomal alterations and encompasses the functional conservation of crucial genomic elements.

Figure 1. Chromosome 13 interactome. A) Complete interactome without clustering. B) Interactome with clustering. The groupings or clusters do not show significant biological associations.

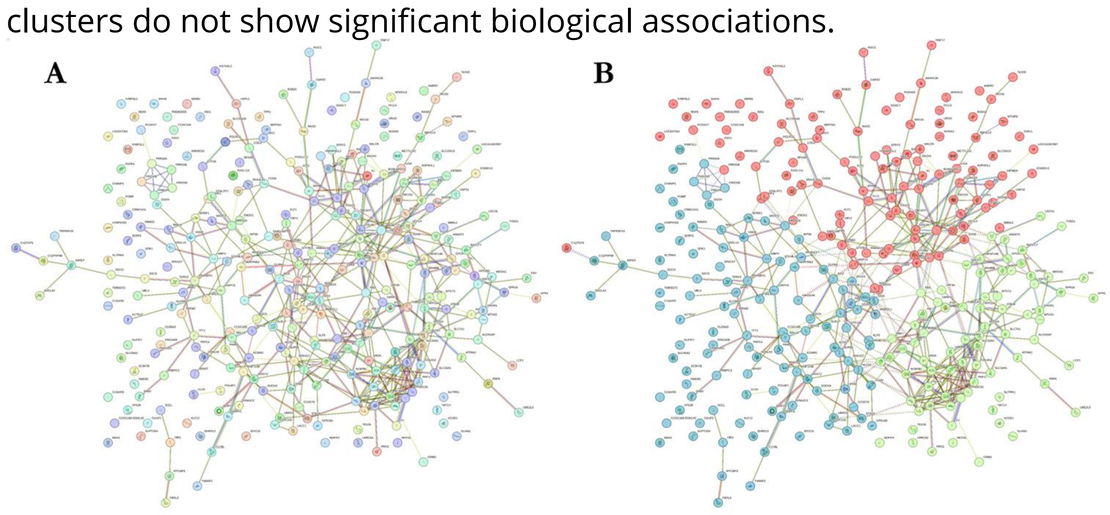
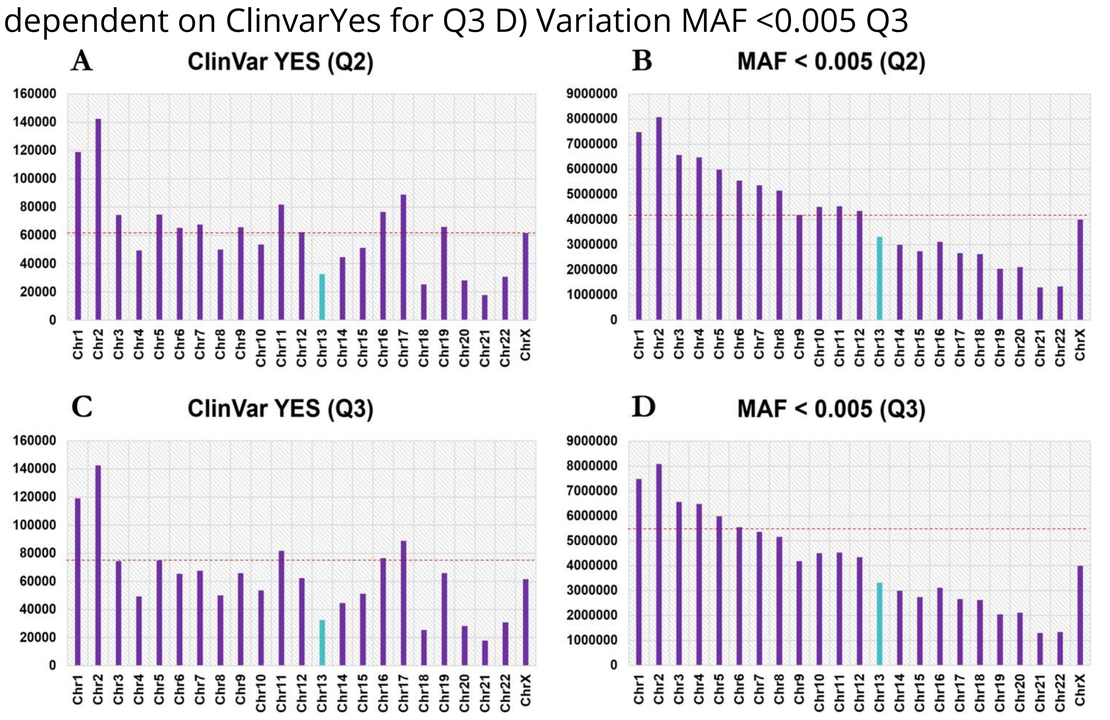


Table 1. Chr13-related diseases based on functional enrichment

Biotools	Diseases	OMIM[a]	p-value	Adjusted p-value[b]
ToppGene	Hemostatic factors and hematological phenotypes	NA	1.049E-4	2.679E-2
	Deafness, autosomal dominant nonsyndromic sensorineural 3	601544	3.102E-4	2.679E-2
	DFNA 3 Nonsyndromic Hearing Loss and Deafness	601544	3.102E-4	2.679E-2
	Deafness autosomal recessive 1A	220290	9.040E-4	2.679E-2
	X-linked mixed hearing loss with perilymphatic gusher	304400	9.040E-4	2.679E-2
Enrichr	Schizophrenia	181500	1.242E-1	5.118E-1
	Aicardi goutieres syndrome	225750	1.514E-1	5.118E-1
	permanent neonatal diabetes mellitus	606176	1.514E-1	5.118E-1
	Ischemic stroke	601367	1.514E-1	5.118E-1
	Malignant tumor of esophagus	133239	1.514E-1	5.118E-1
	neoplasm of the genitourinary tract	142623	1.514E-1	5.118E-1

a)((Online Mendelian Inheritance in Man Identification)); b)((Adjusted p value by Benjamini-Hochberg method))

Figure 2. Analysis by quartiles Q2 and Q3 of present variation on Chromosome 13 concerning other human chromosomes. A) Variation dependent on ClinvarYes for Q2 B) Variation MAF <0.005 Q2 C) Variation dependent on ClinvarYes for Q3 D) Variation MAF <0.005 Q3



CONCLUSIONS

This work provides an in-depth genomic characterization of chr13, highlighting regions of functional and clinical significance. These findings enhance the understanding of chr13's architecture and suggest its influence in regulatory networks and disease mechanisms, offering new insights for therapeutic exploration.

REFERENCES

1.Dunham A, Matthews LH, Burton J, Ashurst JL, Howe KL, Ashcroft KJ, et al. The DNA sequence and analysis of human chromosome 13. Nature [Internet]. 2004;428(6982):522–8. Disponible en: http://dx.doi.org/10.1038/nature02379
2.Chromosome 13. Genet Test [Internet]. 2000;4(1):85–94. Disponible en: http://dx.doi.org/10.1089/109065700316543
3.Hassold T, Hall H, Hunt P. The origin of human aneuploidy: where we have been, where we are going. Hum Mol Genet [Internet]. 2007;16(R2):R203–8. Disponible en: http://dx.doi.org/10.1093/hmg/ddm243
4.Miga KH, Sullivan BA. Expanding studies of chromosome structure and function in the era of T2T genomics. Hum Mol Genet [Internet]. 2021; Disponible en: http://dx.doi.org/10.1093/hmg/ddab214
5.Burns KH. Repetitive DNA in disease. Science [Internet]. 2022;376(6591):353–4. Disponible en: http://dx.doi.org/10.1126/science.abl7399